MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics **WEEKLY CHAPTER TEST – 03 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 6: Triangles** **MM:** 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If $\triangle ABC \sim \triangle DEF$ such that $2AB = DE$ and $BC = 8$ cm, then EF is equal to: [1]

(a) 12 cm (b) 16 cm (c) 4 cm (d) 8 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 16 cm

Given: $\triangle ABC \sim \triangle DEF$, $2AB = DE$ (so $\frac{AB}{DE} = \frac{1}{2}$), and $BC = 8$ cm.

We know that corresponding sides of similar triangles are proportional:

Formula / Theorem Used: $\triangle ABC \sim \triangle DEF \implies \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$

Substituting $\frac{AB}{DE} = \frac{1}{2}$ and $BC = 8$ cm:

$$\frac{1}{2} = \frac{8}{EF} \implies EF = 2 \times 8 = 16 \text{ cm} \implies \boxed{EF = \mathbf{16} \text{ cm}}$$

Q2. A vertical stick 20 cm long casts a shadow 10 cm long on the ground. At the same time, a mobile tower casts a shadow 50 m long. The height of the tower is: [1]

(a) 100 m (b) 120 m (c) 25 m (d) 200 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 100 m

Given: Height of stick = 20 cm = 0.20 m; shadow of stick = 10 cm = 0.10 m; shadow of tower = 50 m. Let height of tower be H m.

At the same time of day, vertical objects and their shadows form similar right triangles (AA similarity):

Formula / Theorem Used: $\frac{\text{Height of Tower}}{\text{Shadow of Tower}} = \frac{\text{Height of Stick}}{\text{Shadow of Stick}}$

Substituting the values in metres:

$$\frac{H}{50} = \frac{0.20}{0.10} = \frac{20}{10} = 2 \implies H = 2 \times 50 = 100 \text{ m} \implies \boxed{\text{Height of tower } = \mathbf{100} \text{ m}}$$

Q3. It is given that $\triangle ABC \sim \triangle DFE$, $\angle A = 30°$, $\angle C = 50°$, $AB = 5$ cm, $AC = 8$ cm, and $DF = 7.5$ cm. Then which of the following is true? [1]

(a) $DE = 12$ cm, $\angle F = 50°$ (b) $DE = 12$ cm, $\angle F = 100°$ (c) $EF = 12$ cm, $\angle D = 100°$
(d) $EF = 12$ cm, $\angle D = 30°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) $DE = 12$ cm, $\angle F = 100°$

Given: $\triangle ABC \sim \triangle DFE$ (vertex correspondence: $A \leftrightarrow D$, $B \leftrightarrow F$, $C \leftrightarrow E$).
We know the angle-sum property of a triangle and similarity properties:

Formula / Theorem Used: $\angle A + \angle B + \angle C = 180°$ and $\dfrac{AB}{DF} = \dfrac{BC}{FE} = \dfrac{AC}{DE}$

Finding $\angle B$ in $\triangle ABC$: $\angle B = 180° - (\angle A + \angle C) = 180° - (30° + 50°) = 100°$.
By vertex correspondence ($B \leftrightarrow F$), $\angle F = \angle B = 100°$.
Finding DE: $\dfrac{AB}{DF} = \dfrac{AC}{DE} \implies \dfrac{5}{7.5} = \dfrac{8}{DE} \implies \dfrac{50}{75} = \dfrac{2}{3} = \dfrac{8}{DE} \implies DE = \dfrac{8 \times 3}{2} = 12$ cm.

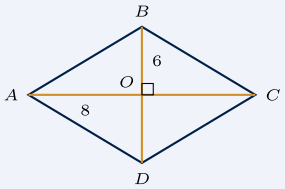
$$\boxed{DE = \mathbf{12}\text{ cm}, \quad \angle F = \mathbf{100°}}$$

Q4. The diagonals of a rhombus are 16 cm and 12 cm. The length of the side of the rhombus is: [1]

(a) 9 cm (b) 10 cm (c) 8 cm (d) 20 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 10 cm



Given: Diagonals of rhombus $ABCD$ are $AC = 16$ cm and $BD = 12$ cm, intersecting at O.
We know that diagonals of a rhombus bisect each other at right angles (90°):

Formula / Theorem Used: $AO = \dfrac{AC}{2}$, $BO = \dfrac{BD}{2}$, and $AB^2 = AO^2 + BO^2$

Here, $AO = \dfrac{16}{2} = 8$ cm and $BO = \dfrac{12}{2} = 6$ cm.
In right-angled $\triangle AOB$ ($\angle AOB = 90°$):

$$AB^2 = 8^2 + 6^2 = 64 + 36 = 100 \implies AB = \sqrt{100} = \boxed{\mathbf{10}\text{ cm}}$$

MATHS MANTHAN • TUITION CLASSES • OFFICIAL SOLUTIONS

Q5. **Assertion (A):** If a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.
Reason (R): This result is known as Basic Proportionality Theorem (Thales' Theorem). [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: A line parallel to one side of a triangle divides the other two sides in the same ratio (proved using equal triangle areas on the same base between the same parallels).

Evaluating (A) & (R): Assertion (A) is the true geometric statement of the theorem, and Reason (R) correctly states its historical name. However, merely naming a theorem does *not* mathematically explain why the ratio equality holds. Hence, both (A) and (R) are true, but (R) is **not** the correct explanation of (A).

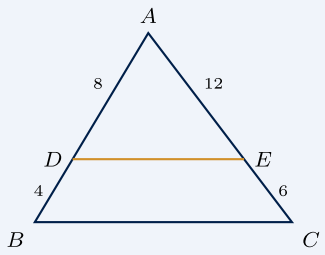
SECTION B: Descriptive Questions MATHS MANTHAN | (1 × 2 + 3 × 3 = 11 Marks)

Q6. In $\triangle ABC$, D and E are points on the sides AB and AC respectively. If $AB = 12$ cm, $AD = 8$ cm, $AE = 12$ cm, and $AC = 18$ cm, show whether $DE \parallel BC$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Finding Remaining Segments DB and EC [0.5 Mark]

Given: $AB = 12$ cm, $AD = 8$ cm, $AE = 12$ cm, $AC = 18$ cm.



$$DB = AB - AD = 12 - 8 = 4 \text{ cm}$$

$$EC = AC - AE = 18 - 12 = 6 \text{ cm}$$

We know the Converse of Basic Proportionality Theorem:

Formula / Theorem Used: If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.

Step 2: Computing & Comparing Ratios [1.5 Marks]

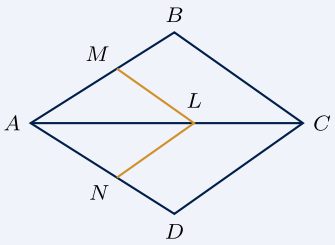
$$\frac{AD}{DB} = \frac{8}{4} = \frac{2}{1} = 2 \quad \text{and} \quad \frac{AE}{EC} = \frac{12}{6} = \frac{2}{1} = 2$$

Since $\frac{AD}{DB} = \frac{AE}{EC} = 2$:

$\therefore$ **DE $\parallel$ BC** (by Converse of Basic Proportionality Theorem)

Q7. In the given figure, if $LM \parallel CB$ and $LN \parallel CD$, prove mathematically that $\dfrac{AM}{AB} = \dfrac{AN}{AD}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: In quadrilateral $ABCD$, L lies on AC, M on AB, N on AD such that $LM \parallel CB$ and $LN \parallel CD$.
To Prove: $\dfrac{AM}{AB} = \dfrac{AN}{AD}$.
We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: If a line is drawn parallel to one side of a triangle intersecting the other two sides, it divides them in the same ratio.

Step 1: Applying BPT in $\triangle ABC$ ($LM \parallel CB$) [1 Mark]
In $\triangle ABC$, since $LM \parallel CB$, by BPT:

$$\frac{MB}{AM} = \frac{LC}{AL}$$

Adding 1 to both sides to form whole sides AB and AC:

$$\frac{MB}{AM} + 1 = \frac{LC}{AL} + 1 \implies \frac{MB + AM}{AM} = \frac{LC + AL}{AL}$$

Since $MB + AM = AB$ and $LC + AL = AC$, taking reciprocals gives:

$$\frac{AB}{AM} = \frac{AC}{AL} \implies \frac{AM}{AB} = \frac{AL}{AC} \quad \text{— (i)}$$

Step 2: Applying BPT in $\triangle ADC$ ($LN \parallel CD$) [1 Mark]
In $\triangle ADC$, since $LN \parallel CD$, by BPT:

$$\frac{ND}{AN} = \frac{LC}{AL} \implies \frac{ND}{AN} + 1 = \frac{LC}{AL} + 1$$

Simplifying and taking reciprocals:

$$\frac{AD}{AN} = \frac{AC}{AL} \implies \frac{AN}{AD} = \frac{AL}{AC} \quad \text{— (ii)}$$

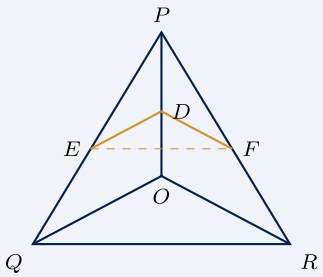
Step 3: Equating Relations (i) and (ii) [1 Mark]
Since both expressions equal $\dfrac{AL}{AC}$:

$$\therefore \frac{AM}{AB} = \frac{AN}{AD} \quad \text{[Hence Proved]}$$

MATHS MANTHAN • OFFICIAL SOLUTIONS

Q8. In the given figure, $DE \parallel OQ$ and $DF \parallel OR$ in $\triangle PQR$. Prove mathematically that $EF \parallel QR$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: O is a point inside $\triangle PQR$ with D on OP, E on PQ, F on PR, such that $DE \parallel OQ$ and $DF \parallel OR$.
To Prove: $EF \parallel QR$.

We know the Basic Proportionality Theorem and its Converse:

Formula / Theorem Used: BPT gives equal ratios from parallel lines; Converse of BPT gives $EF \parallel QR$ when $\dfrac{PE}{EQ} = \dfrac{PF}{FR}$.

Step 1: Applying BPT in $\triangle POQ$ and $\triangle POR$ [2 Marks]

1. In $\triangle POQ$, since $DE \parallel OQ$, by BPT:

$$\frac{PE}{EQ} = \frac{PD}{DO} \quad \text{— (i)}$$

2. In $\triangle POR$, since $DF \parallel OR$, by BPT:

$$\frac{PF}{FR} = \frac{PD}{DO} \quad \text{— (ii)}$$

Step 2: Applying Converse of BPT in $\triangle PQR$ [1 Mark]

From equations (i) and (ii), both left-hand sides equal $\dfrac{PD}{DO}$, so:

$$\frac{PE}{EQ} = \frac{PF}{FR}$$

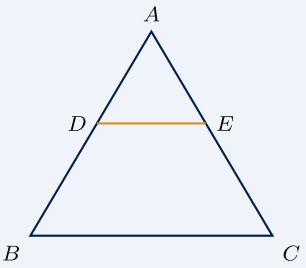
Since line EF divides sides PQ and PR of $\triangle PQR$ in the same ratio, by the Converse of Basic Proportionality Theorem:

$$\boxed{\therefore EF \parallel QR \quad \text{[Hence Proved]}}$$

MATHS MANTHAN • TUITION CLASSES • OFFICIAL SOLUTIONS

Q9. In $\triangle ABC$, D and E are points on sides AB and AC such that $DE \parallel BC$ and $BD = CE$. Prove that: (i) $\triangle ADE$ is an isosceles triangle, (ii) $\triangle ABC$ is an isosceles triangle. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: In $\triangle ABC$, points D and E lie on sides AB and AC respectively such that $DE \parallel BC$ and $BD = CE$.
To Prove: (i) $\triangle ADE$ is an isosceles triangle ($AD = AE$), (ii) $\triangle ABC$ is an isosceles triangle ($AB = AC$).

We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: If $DE \parallel BC$ in $\triangle ABC$, then $\dfrac{AD}{BD} = \dfrac{AE}{CE}$.

Step 1: Proving $\triangle ADE$ is Isosceles [1.5 Marks]

In $\triangle ABC$, since $DE \parallel BC$, by BPT:

$$\frac{AD}{BD} = \frac{AE}{CE}$$

It is given that $BD = CE$. Substituting $CE = BD$ into the denominator on the right-hand side:

$$\frac{AD}{BD} = \frac{AE}{BD}$$

Multiplying both sides by BD ($BD > 0$):

$$AD = AE \qquad \text{— (i)}$$

Since two sides of $\triangle ADE$ are equal ($AD = AE$), $\triangle ADE$ **is an isosceles triangle**.

Step 2: Proving $\triangle ABC$ is Isosceles [1.5 Marks]

Adding the given equality $BD = CE$ to equation (i) ($AD = AE$):

$$AD + BD = AE + CE \implies AB = AC$$

Since two sides of $\triangle ABC$ are equal ($AB = AC$):

$\therefore$ (i) $\triangle ADE$ and (ii) $\triangle ABC$ are isosceles. [Hence Proved]

MANTHAN TUITION CLASSES
MATHS MANTHAN OFFICIAL SOLUTIONS

Q10. Case Study 03: Aeronautical Kite Design & Geometric Proportions

Rahul is designing a traditional diamond kite for an aerodynamics competition. The frame consists of two perpendicular bamboo sticks AC and BD intersecting at O, where AC is the vertical spine and BD is the horizontal crosspiece. In the upper half, $AB = AD = 25$ cm and $OA = 20$ cm. [4]

(i) **[1 Mark]** Find the exact length of line segment OB.

(ii) **[1 Mark]** Which geometric similarity criterion proves that $\triangle AOB \sim \triangle AOD$?

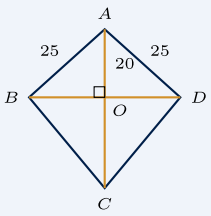
(iii) **[2 Marks]** If a smaller replica is built with scale factor $2 : 5$, find the lengths of the crosspiece and spine of the miniature kite if original $AC = 45$ cm and $BD = 30$ cm.

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Data: Kite frame $ABCD$ has perpendicular sticks $AC \perp BD$ meeting at O ($\angle AOB = \angle AOD = 90°$), $AB = AD = 25$ cm, and $OA = 20$ cm.

We know the Pythagoras Theorem and the similarity scale-factor relation:

Formula / Theorem Used: $\text{Hyp}^2 = \text{Base}^2 + \text{Perp}^2$ and Replica Side $= k \times$ Original Side



Sub-part (i): Finding the Length of OB [1 Mark]

In right-angled $\triangle AOB$ ($\angle AOB = 90°$), applying Pythagoras Theorem:

$$AB^2 = OA^2 + OB^2 \implies 25^2 = 20^2 + OB^2 \implies OB^2 = 625 - 400 = 225 \implies \boxed{OB = \sqrt{225} = \mathbf{15} \text{ cm}}$$

Sub-part (ii): Similarity Criterion Proving $\triangle AOB \sim \triangle AOD$ [1 Mark]

In right-angled $\triangle AOD$ ($\angle AOD = 90°$), by Pythagoras Theorem, $OD = \sqrt{25^2 - 20^2} = \sqrt{225} = 15$ cm $= OB$.

Comparing $\triangle AOB$ and $\triangle AOD$: $\dfrac{AO}{AO} = \dfrac{20}{20} = 1$, $\dfrac{OB}{OD} = \dfrac{15}{15} = 1$, $\dfrac{AB}{AD} = \dfrac{25}{25} = 1$ (also $\angle AOB = \angle AOD = 90°$ included).

$$\boxed{\triangle AOB \sim \triangle AOD \quad \text{by the } \textbf{SSS (or SAS) Similarity Criterion.}}$$

Sub-part (iii): Dimensions of the Miniature Kite Replica (Scale Factor $2 : 5$) [2 Marks]

Given scale factor $k = \dfrac{2}{5}$, original spine $AC = 45$ cm, and original crosspiece $BD = 30$ cm.

$$\text{Miniature Crosspiece } (B'D') = \frac{2}{5} \times 30 = 12 \text{ cm}, \qquad \text{Miniature Spine } (A'C') = \frac{2}{5} \times 45 = 18 \text{ cm}$$

$$\boxed{\text{Miniature Crosspiece } = \mathbf{12} \text{ cm}, \qquad \text{Miniature Spine } = \mathbf{18} \text{ cm}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★